

Materials Science and Technology

ISSN: 0267-0836 (Print) 1743-2847 (Online) Journal homepage: www.tandfonline.com/journals/ymst20

A review of irradiation stability of lithium hydride neutron shielding material

W. Wang, Q. Li, Q. Li, X. Yang & G. Le

To cite this article: W. Wang, Q. Li, Q. Li, X. Yang & G. Le (2016) A review of irradiation stability of lithium hydride neutron shielding material, Materials Science and Technology, 32:5, 434-437, DOI: 10.1179/1743284715Y.0000000105

To link to this article: https://doi.org/10.1179/1743284715Y.0000000105

Published online: 03 Mar 2016.

Submit your article to this journal

Article views: 622

View related articles

View Crossmark data

Citing articles: 6 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=ymst20

A review of irradiation stability of lithium hydride neutron shielding material

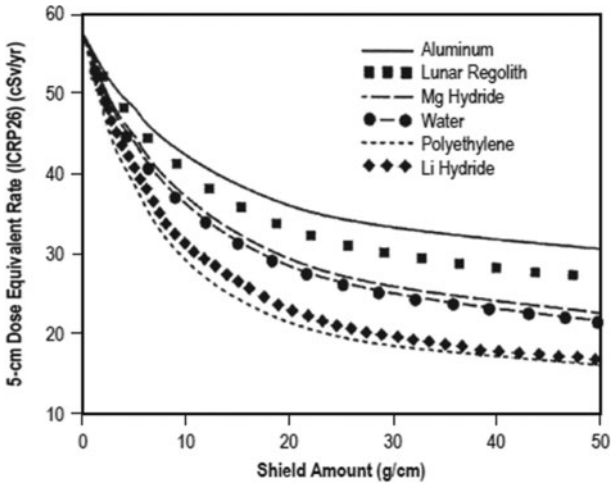
W. Wang*, Q. Li, Q. Li, X. Yang and G. Le

Lithium hydride remains theoretically the most efficient neutron shield material, showing immense application prospects. The properties of Lithium hydride neutron shielding material, such as use temperature, mechanical properties, thermal conductivity and irradiation swelling behaviour, were reviewed. Factors that affect the irradiation swelling of Lithium hydride were also discussed in this paper. The development trends of lithium hydride neutron shielding material were proposed.

Keywords: Lithium hydride, Neutron shielding, Irradiation swelling

With the rapid development of nuclear energy and nuclear reactors, advanced neutron shielding materials for radiation protection are highly demanded. Especially in the application of nuclear reactor, the quality and volume of neutron shielding material account for a large proportion, which greatly hinder the development of nuclear technology. Therefore, it is urgent to obtain optimised neutron shielding materials that have excellent shielding performance, good heat resistance and light weight.

Lithium hydride is the best neutron moderator and absorbent by far and has been selected by the Aircraft Nuclear Propulsion (ANP) as a primary shielding material because of the combination of high hydrogen density ($\sim 5.87 \times 10^{22}$ H atoms cc^{-1}), high melting point ($\sim$956 K), low density ($\sim$0.775 g cc^{-1}), high absorption cross-section and low molecular weight, all of which are ideal for use as a low mass, high efficiency neutron shield.[1] Naval Research Laboratory compared the radiation exposure rate for various materials as a function of their thickness in mass per unit area.[2] The results showed that the shielding effectiveness of lithium hydride and polyethylene was excellent, as shown in Fig 1. The performance of LiH would be better because the stability of polyethylene was poor. Knolls Atomic Energy lab of America also conducted a research on the space of shielding materials in the Plo Michel J plan in 2006. Eighty-eight kinds of materials such as B_4C, H_2O and lithium hydride were investigated. Lithium hydride was found to be the most competitive material. To date, the United States of America has investigated the use of lithium hydride as shielding material in three major projects: the ANP Program, Systems for Nuclear Auxiliary Power and the SP-100 Program. The Union of Soviet Socialist Republics adopted at least 30 TOPAZ-1 systems that using lithium hydride reactor shields. However, the inherent properties of lithium hydride such as mechanical, thermal and irradiation stability were poor, which limited its application.[3–4] In order to meet the needs of neutron shielding material, USA and Russia have began to investigate the characteristics of lithium hydride and optimise its properties especially irradiation stability since 1940s and have made breakthrough progress.[5–12]



1 Dose at 5 cm depth in tissue for GCRs at solar minimum as function of areal density for various materials

In this paper, the neutron shielding properties especially the irradiation swelling performance of lithium hydride are reviewed, the possible reasons and solving method for irradiation swelling of lithium hydride are analysed and the trends of lithium hydride neutron shielding material is also pointed out.

Properties of lithium hydride material

Use temperature of lithium hydride under irradiation

The melting temperature of lithium hydride is high ($T_m = 956$ K); however, it begins to decompose at $\sim$680 K, as shown in equation (1).

$$2LiH \rightarrow 2Li + H_2(gas) \quad (1)$$

The reaction is accelerated with increasing temperature. A large amount of hydrogen will be generated at high temperature and accumulated in the body, which causes volume expansion. After expanding to a certain extent,

Institute of Materials, China Academy of Engineering Physics, PO box 919-71, Mianyan 621900, Sichuan, China

*Corresponding author, email 13628080837@163.com

© 2016 Institute of Materials, Minerals and Mining
Published by Taylor & Francis on behalf of the Institute
Received 06 January 2015; accepted 17 June 2015
DOI 10.1179/1743284715Y.0000000105

the capsule will leak or rupture. Hence, the use temperature of lithium hydride is limited to 600–800 K to prevent unacceptable swelling, especially for a high dose region.[13–14]

Thermal conductivity of lithium hydride

The heat transfer efficiency will be enhanced with improvement of the thermal conductivity of lithium hydride, which reduces the temperature gradient and improves irradiation stability of the shielding. National Aeronautics and Space Administration, Naval Research Laboratory and Oak Ridge National Laboratory studied the thermal conductivity of lithium hydride and found that temperature had significant effects on thermal conductivity. With the increasing of temperature, the thermal conductivity of lithium hydride decreased from 11.06 W mK^{-1} ($T = 200$ K) to 3.5 W mK^{-1} ($T = 350$ K). An addition of Li will increase the thermal conductivity by adding additional free electrons. Since conducting electrons are dominant methods of energy transport at elevated temperatures, the thermal conductivity decreases as temperature increases.[15]

Mechanical properties of lithium hydride

Welch studied the ultimate tensile strength (UTS) of cold pressed and sintered lithium hydride. The UTS of lithium hydride as a function of temperature is given in Table 1. The tensile strength of lithium hydride is 17.17–19.55 MPa at room temperature. With the increase in temperature, the tensile strength decreases to 1.71–2.73 MPa ($T = 728$ K), and Young's modulus also decreases from 4000–6000 MPa ($T = 339$ K) to 160–210 MPa ($T = 689$ K).[16]

Irradiation stability

Potential irradiation swelling mechanisms

The swelling of lithium hydride would occur at radiations such as X-rays, gamma rays, neutrons, β, low energy electrons and even ultraviolet.[17–21] Irradiated lithium hydride as determined by dimensions and mass was studied, and it showed three distinct phases at 323 K and above: region I – rapid initial growth; II –slower, near linear swelling; III – erratic and unpredictable growth with eventual sample breakup.[22] Accompanying the swelling is the formation of aggregates of metallic lithium and gaseous hydrogen within the lattice. The state of these decomposition products has been characterised. It has been observed that hydrogen present in the lattice can be converted to a 'motionally narrowed' gaseous form by radiation. This gross decomposition is associated with volume changes of 25 per cent or more.[23–26] Generally, hydrogen gas (H_2) is believed to form due to irradiation stripping of electrons off adjacent H atoms in the lithium hydride crystal lattice, allowing the atoms to bond to each other, forming relatively stable H_2. The formed H_2 either stays in position, or diffuses to form bubbles (or leaves the sample), or recombines with Li to form lithium hydride. Newkirk and Leider[27–28] researched the structure of irradiated lithium hydride and found that Li precipitated for lithium hydrogen at high temperature ($T \geqslant 150$°C). Cubic hydrogen bubbles were also observed in irradiated single lithium hydrogen crystals,[29–33] which proved the hypothesis above.

Table 1 Ultimate tensile strength as function of temperature for lithium hydride

T/K	UTS/MPa
298	17.17–19.55
533	13.21–13.72
644	5.72–6.00
728	1.71–2.73

The mechanisms that affect the irradiation stability of the shield design would be (i) the formation of the H_2, (ii) the migration of H_2 to form bubbles (swelling) and (iii) the recombination of H and Li to form lithium hydride (reducing swelling). If the rate of recombination exceeds the rate of formation of the H_2 bubbles, the swelling would be reduced or minimised. If there is LiOH inside lithium hydride specimen, it will react with lithium hydride to form additional H_2. The kinetics are expected to be dictated by temperature and the proximity of the reactant phases.

Factors affecting the irradiation swelling of the lithium hydride

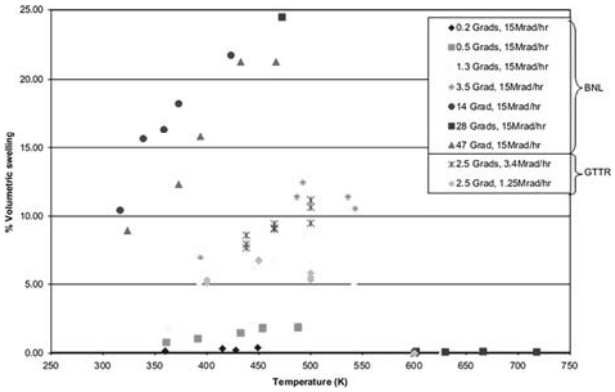
Irradiation stability of lithium hydride is the major problem for its application as radiation shielding materials. Reports showed that the body swelling in part of lithium hydride specimen was up to 25% under gamma ray irradiation.[13] The large swelling resulted in a breakage of lithium hydride, which largely shortened the service life of the lithium hydride. To improve the irradiation stability of lithium hydride, much work focused on the behaviours of irradiated swelling of lithium hydride has been carried out by USA and Russia. Naval Reactors Prime Contractor Team believes that the potential key issues affecting lithium hydride swelling are (i) radiation type: gamma only, neutron only (not discussed) or mixed neutron and gamma; (ii) neutron spectrum: fast, epithermal and thermal; (iii) temperature; (iv) dose rate; (v) dose; (vi) LiOH impurity level; (vii) processing method (cast, cold pressed, cold pressed + sintered and hot pressed); (viii) stoichiometry; (IX) thermal migration – as a function of stoichiometry; (X) environment (hydrogen overpressure).

Dose and dose rate

Brookhaven National Laboratory and Georgia Institute of Technology Research Reactor performed the gamma and mixed neutron and gamma irradiation testing of lithium hydride.[16] The results showed that higher total dose and dose rate resulted in higher volumetric swelling and swelling drops off to $<1\% > 600$ K, as shown in Fig. 2. When the irradiation dose of gamma ray is low (<0.5 Grads), the swelling of lithium hydride is small (body swelling $<2\%$), and when the irradiation dose is higher than 1.3 Grads, the irradiation swelling increased significantly. With the increase in dose rate, the swelling of lithium hydride also significantly increased. Minushkin performed a series of irradiations to determine the stability of lithium hydride in a radiation environment and found that lithium hydride samples swell, turn into powder and finally flake with the increase in neutron/dose from 2×10^{16} to 1.12×10^{18} Mrad h^{-1}.

Temperature

Irradiation swelling of lithium hydride is significantly affected by temperature. Pretzel[34–35] has established that



2 Brookhaven National Laboratory and GTRR gamma test results on lithium hydride[16]

ionising radiation generate F centres and V centres in lithium hydride crystals without obvious swelling when they are cooled to liquid nitrogen temperature during irradiation. However, irradiation swelling of lithium hydride becomes significant at room temperature or above, and the effect of temperature on swelling is not monotonic. When the temperature is below $\sim$160°C, under the same dosage of ^{60}Co gamma irradiation, irradiation swelling increases with elevating of temperature. When temperature is between 160 and 200°C, swelling reaches the maximum (body swelling up to 25%). However, the swelling almost disappeared at temperatures $>$300°C,[36] and the mechanism is not clear. The swelling behaviour of lithium hydride in the gamma/neutron irradiation has also been studied. The results showed that the swelling achieved maximum when temperature was between 400 and 600 K, and almost disappeared when temperature was 800 K.

Impurities

The content of impurities especially LiOH have great influence on the swelling behaviours of lithium hydride. Dinh[37–38] believes that LiOH reacts with lithium hydride to produce hydrogen gas (equation (2)[39–40]). When hydrogen bubble is small, bubbles gather or absorb more H_2 due to the surface tension effect, causing the internal pressure of lithium hydride to increase. When the pressure inside the bubble is greater than surface tension, microcrack or rupture will occur.

$$LiOH + LiH \rightarrow Li_2O + H_2(gas) \quad (2)$$

Preparation process

About 47 tests of SNAP and ANP neutron irradiations were reported to select neutron shielding material. The results showed that the influence of preparation process of lithium hydride on the irradiation swelling was significant.

Cast specimens before irradiation produced less swelling and evolved less gas than cold pressed specimens that were not outgassed. This behaviour might be explained as follows. First, the melting and zone refinement likely convert any existing LiOH into Li_2O, and move it to the surfaces. If the reaction as equation (1) happens, the hydrogen will be ready to escape. Second, casting is performed under an inert atmosphere (likely hydrogen) with very little if any humidity, which limits LiOH formation. Subsequent to forming the specimen by casting, LiOH can form. Then, upon heating, the LiOH will simply evaporate. In the end, casting can create a different microstructure than cold pressing, which may facilitate the removal of any hydrogen build-up within the specimen, and this requires further research.

Outgassed specimens before irradiation are less likely to swell. It is not clear how this works. On a simple level, gas might be removed from the lithium hydride and therefore is no longer present to expand and drive swelling. Perhaps more likely, however, outgassing removes LiOH, which is believed to evolve hydrogen through a chemical reaction with lithium hydride. The evolved hydrogen can contribute to the swelling during irradiation if it is trapped within the lithium hydride. The process by which LiOH is removed is given in equation (2).

Hydrogen migration within lithium hydride

The advantage of lithium hydride as a neutron shield material is dependent on the total amount of hydrogen and the uniformity of the hydrogen distribution. When lithium hydride is used as a shielding material, the temperature of lithium hydride near the neutron source will be higher than the other end due to neutron collisions and other reasons caused by the thermal radiation. The thermal gradient induced by the reactor (along the axis of the shield) will cause the hydrogen to migrate to the colder regions of the shield. The region closest to the reactor becomes less effective, allowing neutrons to penetrate further into the shield, which will reduce the neutron shielding performance and make it more prone swelling.[41]

Welch[42] performed hydrogen migration studies, which concluded that adding 2% free lithium will not reduce the diffusion rate of H_2, but will increase the thermal diffusion of lithium hydride obviously. This would reduce the temperature difference between the materials, reduce the migration of the hydrogen atom, improve composition uniformity and reduce irradiation swelling.

Conclusions

Lithium hydride remains theoretically the most efficient neutron shield material. However, it might be swelling in irradiation environment. The swelling could be eased through decreasing the content of LiOH, optimising preparation progress and improving thermal conductivity, and the mechanism should be studied systemically. Moreover, with sufficient testing and development, the properties of lithium hydride can be optimised for application in future flights. Key points in future research are as follows.

(i) Discovering swelling mechanism of lithium hydride through studying microstructure, chemical composition and surface properties of lithium hydride during irradiation.

(ii) Confirming the key factors that affect irradiation swelling of lithium hydride, finding out how it works and mastering lithium hydride neutron shielding material preparation methods.

Acknowledgement

This work was sponsored by the National Natural Science Foundation of China (grant no. 11305149).

References

1. F. H. Welch: 'Lithium hydride: a space age shielding material', *Nucl. Eng. Des.*, 1974, **26**, 444–460.
2. J. H. Adams Jr, D. H. Hathaway, R. N. Grugel and J. W. Watts: 'Revolutionary concepts of radiation shielding for human exploration of space', *Nasa Tm*, 2005, 4–6.
3. C. E. Johnson, R. R. Heinrich and C. Crouthamel: 'Thermodynamic properties of lithium hydride by an electromotive force method', *J. Phys. Chem.*, 1966, **70**, (1), 242–246.
4. J. B. Condon: 'Preparation and properties of lithium hydride', *ChemInform*, 1992, **23**, (4), 34–42.
5. R. C. Bowman, Jr: 'Distribution of helium in metal tritides', *Nature*, 1978, **271**, 531–533.
6. T. A. Dellin: 'Low-temperature volume expansion of LiH:LiT', *Phys. Rev. B*, 1970, **1B**, (4), 1745–1753.
7. Ya. O. Shablovskii: 'The low-temperature thermodynamic properties of lithium hydride isotopomers', *Russ. J. Phys. Chem. A*, 2010, **84**, (2), 203–208.
8. T. N. Melnikva and K. A. Yakimovich: 'Thermal preperties of lithium hydride and its isotopic modifications in a crystalline state', *High Temp.*, 1980, **18**, (2), 250–256.
9. Y. W. Kim, R. Kaplan and P. J. Bray: 'Electron-spin paramagnetic resonance studies of metallic lithium in neutron-irradiated LiF', *Phys. Rev.*, 1960, **117**, (3), 740–742.
10. M. Belline, P. Denatale and M. Inguscio: 'Laboratory measurements of rotational transitions of lithium hydride in the far-infrared', *Astrophys. J.*, 1994, **424**, 507–509.
11. F. E. Pretzel and R. L. Petty: 'X-ray study of interstitial defects in radiation-damaged alkali halides and lithium hydride', *Phys. Rev.*, 1962, **127**, 777–784.
12. J. J. Gilman and W. G. Johnston: 'Dislocations, point defect clusters, and cavities in neutron irradiated LiF crystals', *J. Appl. Phys.*, 1958, **29**, 877–888.
13. P. C. Sourers, C. L. Folkers, T. Mccreary, G. D. Snider, F. Vanderhoofven and R. T. Tsugawa: 'Swelling and outgassing of tritiated lithium hydride compacts', *J. Nucl. Mater.*, 1989, **168**, 1–11.
14. J. A. Emig, S. R. Mayhugh, F. Garcia and P. C. Souers: 'Swelling of tritiated lithium hydride over the first 150 days', *J. Nucl. Mater.*, 1990, **173**, (2), 113–117.
15. L. B. Fieldhouse: 'Measurements of thermal properties', Wright Air Development Center Report WADC-TR-274; Armour Research Foundation, Chicago, IL. .
16. F. H. Welch: 'Lithium hydride technology I. Properties of lithium hydride and corrosion studies', Atomics International Document NAA-SR-9400; 1964.
17. R. C. Bowman: 'EPR of ultraviolet irradiated lithium hydride crystals', *Solid State Commun.*, 1972, **11**, 1489–1493.
18. W. Burton Lewis and F. E. Pretzel: 'Properties of lithium hydride-III. Paramagnetic resonance of color centers', *J. Phys. Chem. Solids*, 1961, **19**, 139–146.
19. C. Haertling, I. Usov and Y. Wang: 'Outgassing from alpha particla irradiation of lithium hydride and lithium hydroxide', *Nucl. Instrum. Methods Phys. Res. B*, 2011, **269B**, 444–451.
20. P. C. Souers, T. S. Blake, R. M. Penpraze and C. Cline: 'Pulsed nuclear magnetic resonance of gamma-irradiated lithium hydride', *J. Phys. Chem. Solids*, 1969, **30**, 2649–2656.
21. B. V. Shulgin, F. F. Gavilov, B. L. Dvinyaninov, V. I. Koryakov and A. K. Chirkov: 'Spin resonance of irradiated luminesent LiH crystals', *Sov. Phys. J.*, 1967, **10**, (1), 37–38.
22. F. E. Pretzel, W. B. Lewis, E. G. Szklarz and D. T. Vier: 'Properties of lithium hydride V vacancy formation, cavitation, and lithium precipitation in irradiated lithium hydride', *J. Appl. Phys.*, 1962, **33**, 510–518.
23. P. C. Souers, C. L. Folkers, T. Mccreary, R. O. Lindahl, G. D. Snider, F. Vanderhoofven and R. T. Tsugawa: 'Swelling of tritiated lithium hydride powder', *J. Nucl. Mater.*, 1990, **173**, 159–163.
24. F. F. Gavrilov, N. I. Kanunnikov, E. G. Kovalenko, G. I. Pilipenko, N. I. Silkin and V. G. Stepanov: 'Conduction electron paramagnetic resonance of colloidal lithium in ultraviolet irradiated', *Sov. Phys. J.*, 1974, **17**, (6), 880–882.
25. G. S. Smith: 'X-ray diffraction studies of irradiated lithium hydride: oriented lithium precipitates', *J. Phys. Chem. Solids*, 1975, **36**, (7-8), 797–799.
26. C. D. Knutson, H. O. Hooper and P. J. Bray: 'A nuclear magnetic resonance study of decomposition in neutron-irradiated LiF', *J. Phys. Chem. Solids*, 1966, **27**, 147–161.
27. H. W. Newkirk and H. R. Leider: 'Thermal annealing of irradiatied lithium hydride-I. Measurement of stored energy', *J. Phys. Chem. Solids*, 1970, **31**, 1069–1073.
28. H. R. Leider: 'Thermal annealing of irradiated Lithium Hydride-II: A Proposed mechanism', *J. Phys. Chem. Solids*, 1970, **31**, (6), 1221–1224.
29. M. Toshikatsu and T. Motoji: 'Thermoluminescence of LiH Irradiated at low temperature', *J. Phys. Soc. Jpn.*, 1982, **51**, (9), 2862–2868.
30. R. C. Bowman, Jr and A. Attalla: 'Density of trapped gas in heavily-irradiated lithium hydride', *J. Nucl. Mater.*, 1988, **154**, 318–331.
31. F. E. Pretzel and R. L. Petty: 'Interstitial defects in LiH and NaCl irradiated at low temperatures', *Phys. Rev.*, 1961, **123**, (2), 464–465.
32. T. Imaia: 'Observation of bubbles in gamma-irradiated lithium hydride', *Philos. Mag.*, 1970, **21**, (170), 281–286.
33. P. C. Souers, T. Imai, T. S. Blake, R. M. Penpraze and H. R. Leider: 'NMR and electron microscopy studies on irradiated lithium hydride', *J. Phys. Chem. Solids*, 1970, **31**, 1461–1466.
34. F. E. Pretzel, G. V. Gritton, C. C. Rushing, R. J. Friauf, W. B. Lewis and P. Waldstein: 'Properties of lithium hydride-IV: F-centre formation at low temperatures', *J. Phys. Chem. Solids*, 1962, **23**, 325–337.
35. P. C. Sourers, T. A. Jolly and C. F. Cline: 'Pulsed nuclear magnetic resonance study of irradiated lithium hydride', *J. Phys. Chem. Solids*, 1967, **25**, 1717–1719.
36. P. C. Sourers, C. L. Folkers, T. Mccreary, R. O. Lindahl, G. D. Snider, F. Vanderhoofven and R. T. Tsugawa: 'Swelling of tritiated lithium hydride powder', *J. Nucl. Mater.*, 1990, **173**, 159–163.
37. L. N. Dinh, W. McLean II, M. A. Schildbach, J. D. LeMay, W. J. Siekhaus and M. Balooch: 'The nature and effects of the thermal stability of lithium hydroxide', *J. Nucl. Mater.*, 2003, **317**, 175–188.
38. L. N. Dinh, D. M. Grant, M. A. Schibach, R. A. Smith, W. J. Siekhaus, B. Balazs, J. H. Lechey, J. R. Kirkpatrick and W. McLean II: 'Kinetic measurement and prediction of the hydrogen outgassing from the polycrystalline LiH/Li2O/LiOH system', *J. Nucl. Mater.*, 2005, **347**, 31–43.
39. C. Haertling, R. J. Hanrahan, Jr and R. Smith: 'A literature review of reactions and kinetics of lithium hydride hydolysis', *J. Nucl. Mater.*, 2006, **349**, 195–233.
40. M. Balooch, L. N. Dinh and D. F. Calef: 'The reaction kinetics of lithium salt with water vapor', *J. Nucl. Mater.*, 2002, **303**, 200–209.
41. D. M. Charles: 'Fabrication and properties of lithium hydride', *Nucl. Eng. Des.*, 1973, **25**, 309–314.
42. F. H. Welch: 'Lithium hydride technology IV. A novel neutron-gamma material for SNAP shielding applications', Atomics International Document NAA-SR-9400 Volume IV, 1967.